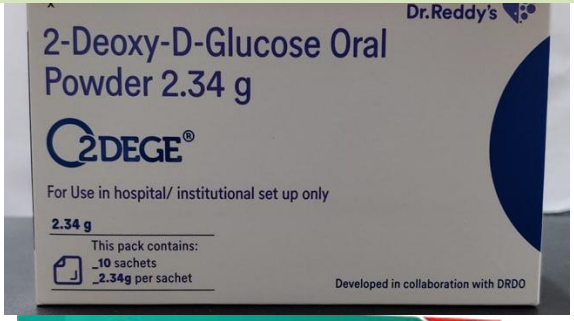
Synthesis of some common medicines

School of Basic Sciences
Indian Institute of Technology, Bhubaneswar

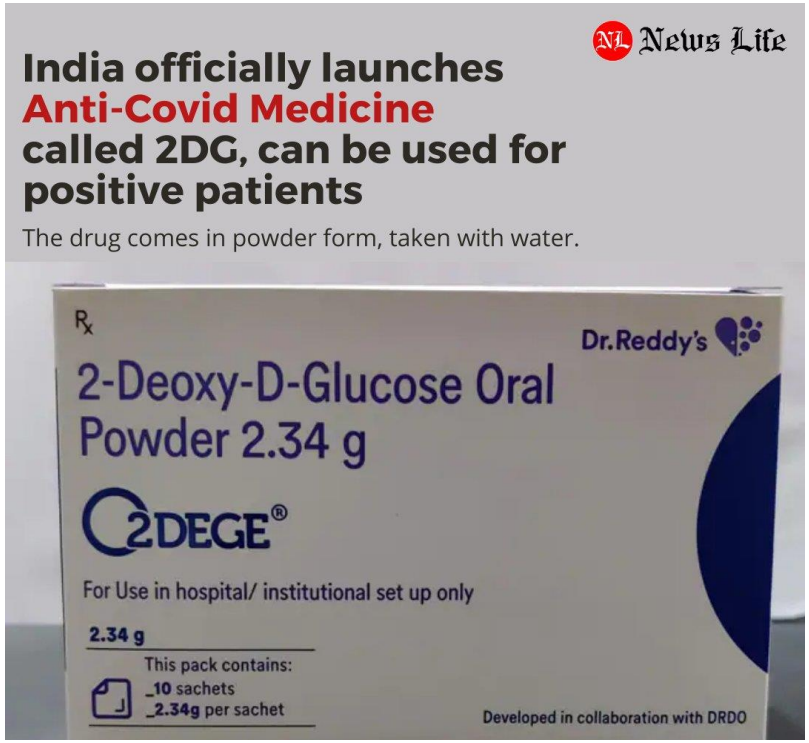
Contents



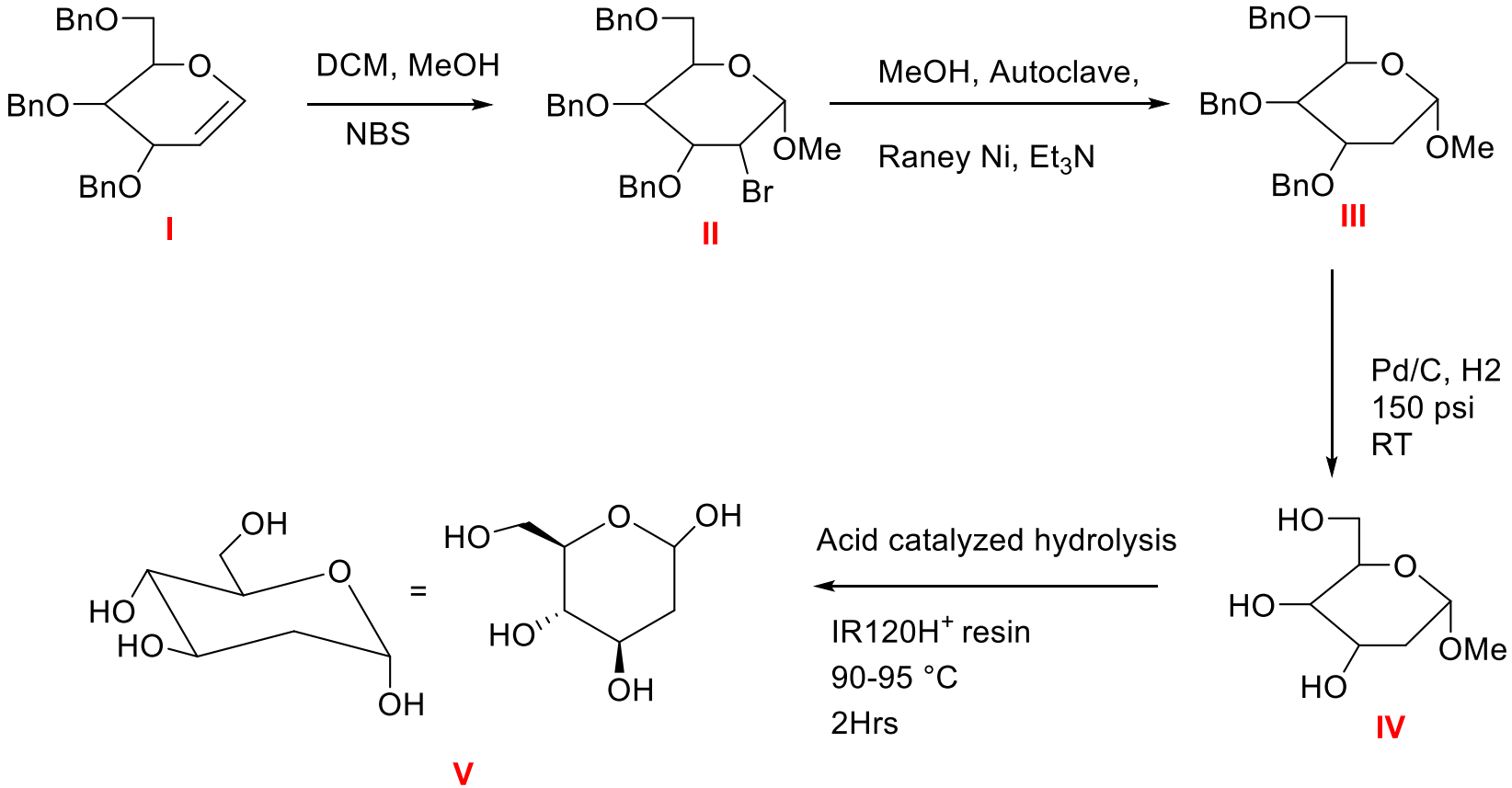
Industrial Chemistry: **Industrial organics with emphasis to Drugs**

2-deoxy-D-glucose (2DG)

- In May, 2021, the Drugs Controller General of India (DCGI) approved an oral formulation of 2-deoxy-D-glucose (2DG) for emergency use as adjunct therapy in moderate to severe coronavirus patients.
- The drug was developed by the DRDO along with Dr. Reddy's Laboratories.



Synthesis of 2-deoxy-D-glucose (2DG) :



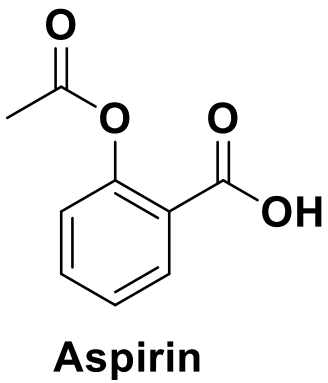
3,4,6-tri-O-benzyl-D-glucal (I)
methyl 2-bromo 2-deoxy-3,4,6-tri-O-benzyl-α/β-D-gluco-/mannopyranoside (II)
methyl 2-deoxy-3,4,6-tri-O-benzyl-α/β-D-gluco-/mannopyranoside (III)
methyl 2-deoxy-α/β-D-glucopyranoside (IV)
2-deoxy-D-glucose (V)

Reference: US patent: US 6,933,382 B2

https://newdrugapprovals.org/2021/05/08/2-deoxy-d-glucose/

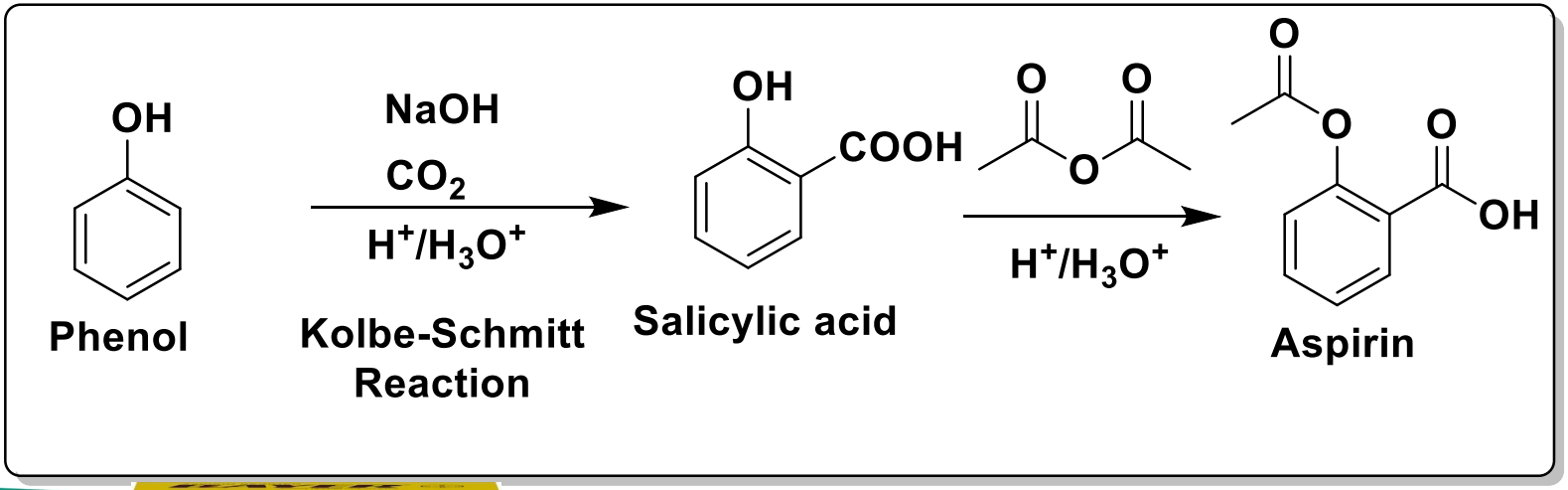
Aspirin

- Aspirin is used to reduce pain, fever or inflammation and lowering the risk of heart attack of clot related strokes.

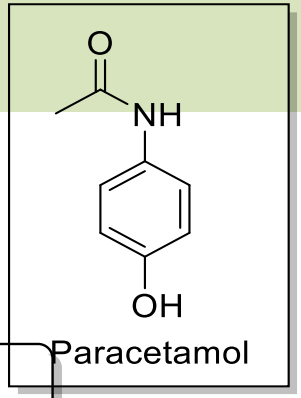


Aspirin

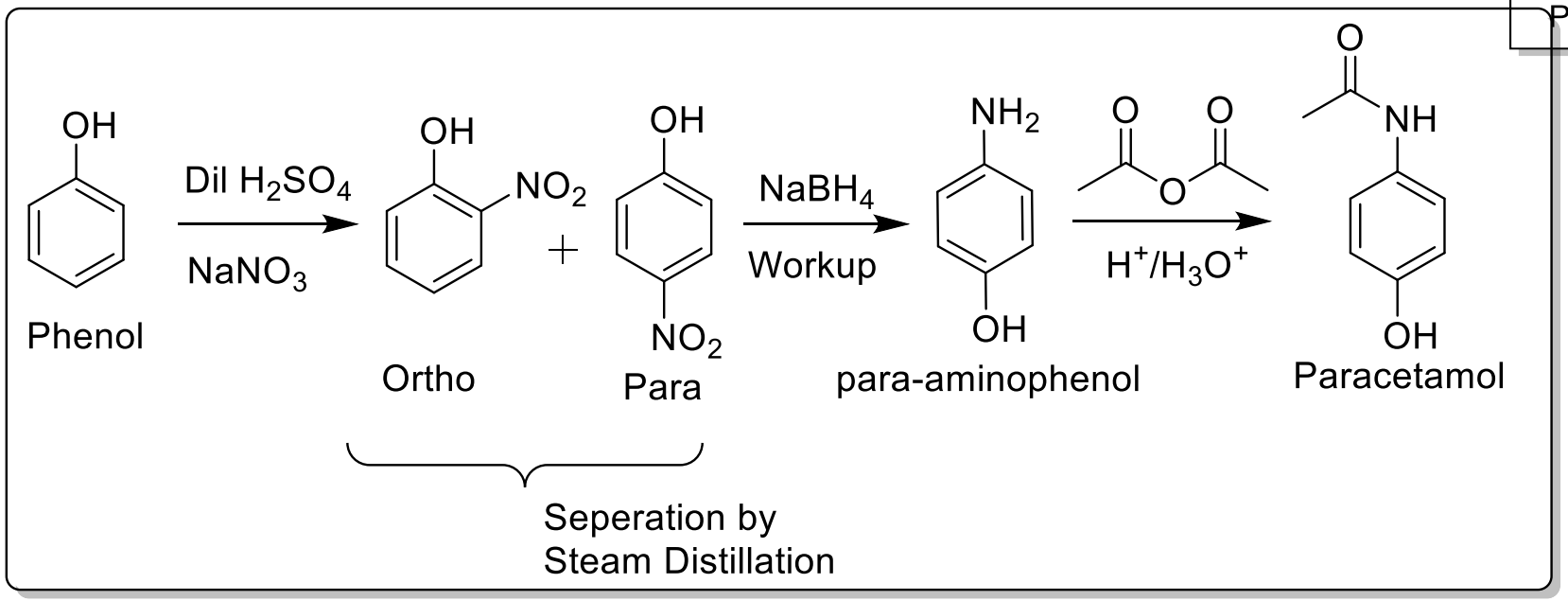
2-Acetoxybenzoic acid (IUPAC name)
or
acetylsalicylic acid



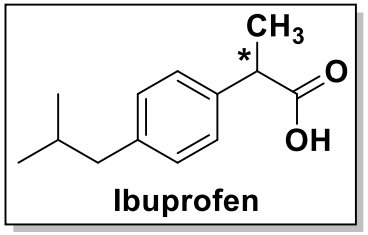
Paracetamol

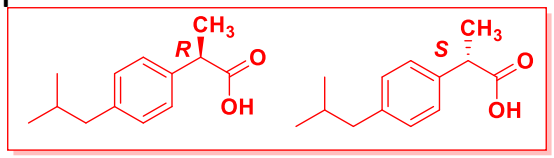


- Paracetamol is also known as acetaminophen. It is a medicine used to treat pain and fever.

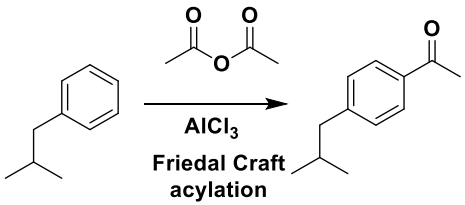


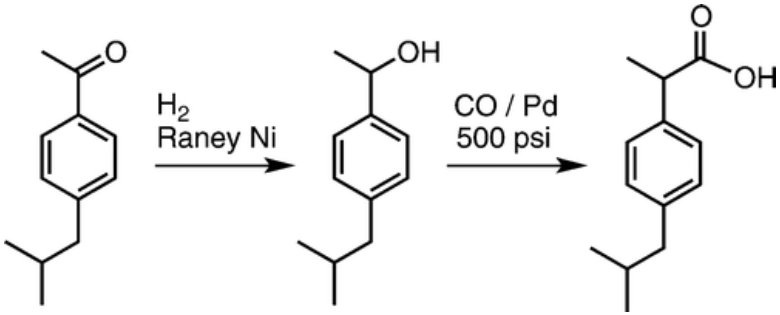
Ibuprofen



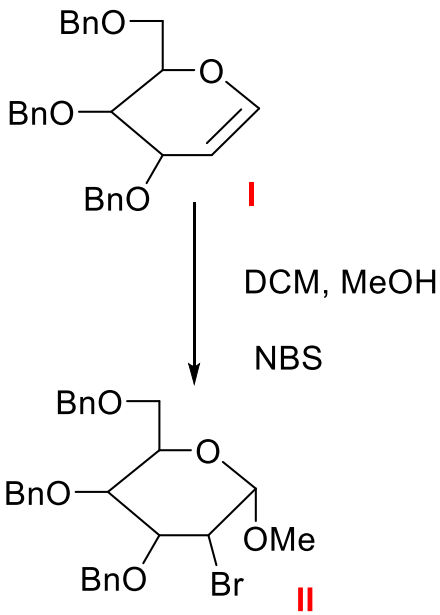


- Ibuprofen is used as a medicine in the nonsteroidal anti-inflammatory drug (NSAID) class that is used for treating pain, fever and inflammation.
- This includes painful migraines, and rheumatoid arthritis.





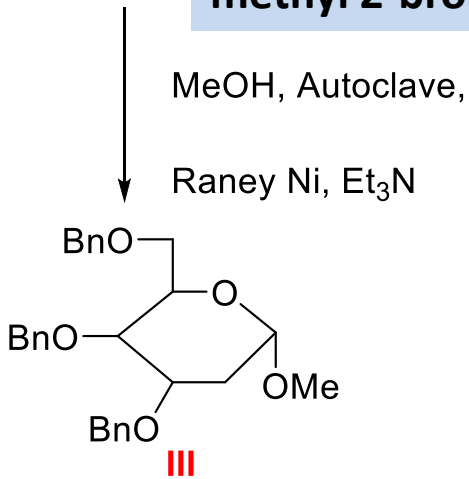
Scheme. The BHC Procedure for Making Ibuprofen on an Industrial Scale



3,4,6-tri-O-benzyl-D-glucal (I)

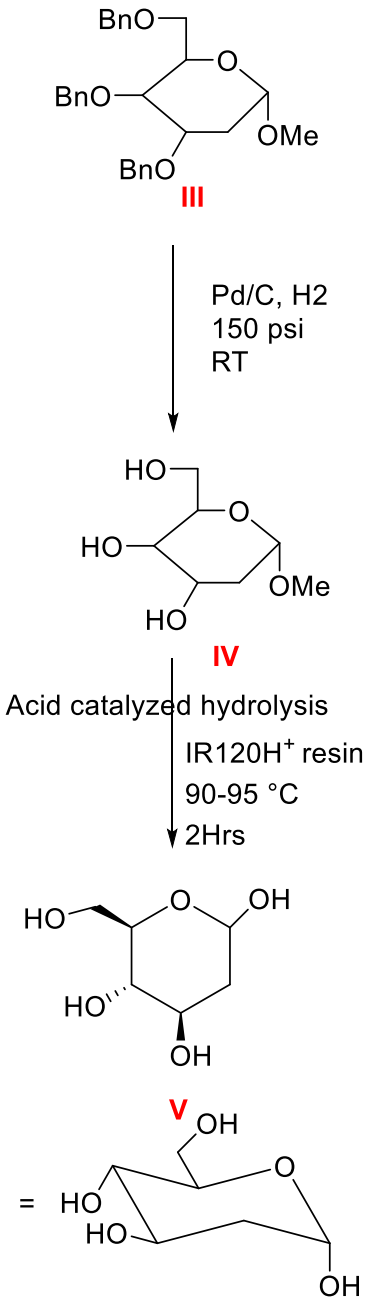
To a solution of 3,4,6-tri-O-benzyl-D-glucal (39 g, 0.09 mmol) in dichloromethane (20 ml) and methanol (100 ml) was added N-bromosuccinimide (18.7 g, 0.09 mil) during 10 min. at room temperature and stirred for 4 h. After completion of the reaction solvent was distilled off. The resultant residue extracted into carbon tetrachloride (2×100 ml) and organic phase concentrated to obtain methyl 2-bromo 2-deoxy-3,4,6-tri-O-benzyl-α/β-D-gluco-/mannopyranoside as a syrup

methyl 2-bromo 2-deoxy-3,4,6-tri-O-benzyl-α/β-D-gluco-/mannopyranoside (II)



A solution of methyl 2-bromo-2-deoxy-3,4,6-tri-O-benzyl/α/β-D-gluco-/mannopyranoside (50 g) in methanol (300) was charged into one liter autoclave along with Raney nickel (10 ml) Et_3N (135 ml) and subjected to hydrogenation at 120 psi pressure at 50° C. for 8 h. After completion of the reaction the catalyst was filtered off and the residue washed with methanol (25 ml). The filtrate was concentrate to obtain methyl 2-deoxy-3,4,6-tri-O-benzyl-α/β-D-glucopyranoside as a syrup (37.9 g, 89%)

methyl 2-deoxy-3,4,6-tri-O-benzyl-α/β-D-gluco-/mannopyranoside (III)



methyl 2-deoxy-3,4,6-tri-O-benzyl-α/β-D-gluco-/mannopyranoside (III)

The syrup of methyl 2-deoxy-3,4, 6-tri-O-benzyl-α/β-D-glucopyranoside (37.9 g) was dissolved in methanol (200 ml). 1 g of 5% Pd/C was added and hydrogenated at 150 psi pressure at room temperature. After 5 hours catalyst was filtered off and solvent evaporated. Quantity of the methyl 2-deoxy-α/β-D-glucopyranoside obtained 10.5 g (70%).

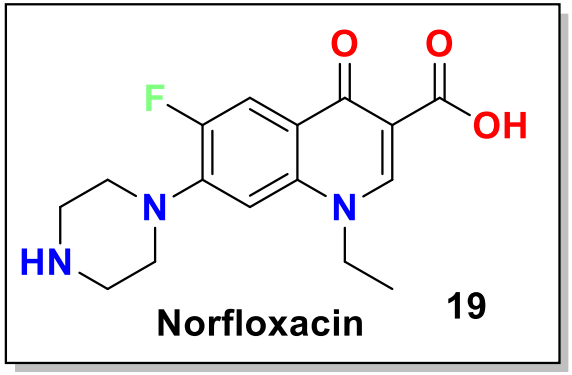
methyl 2-deoxy-α/β-D-glucopyranoside (IV)

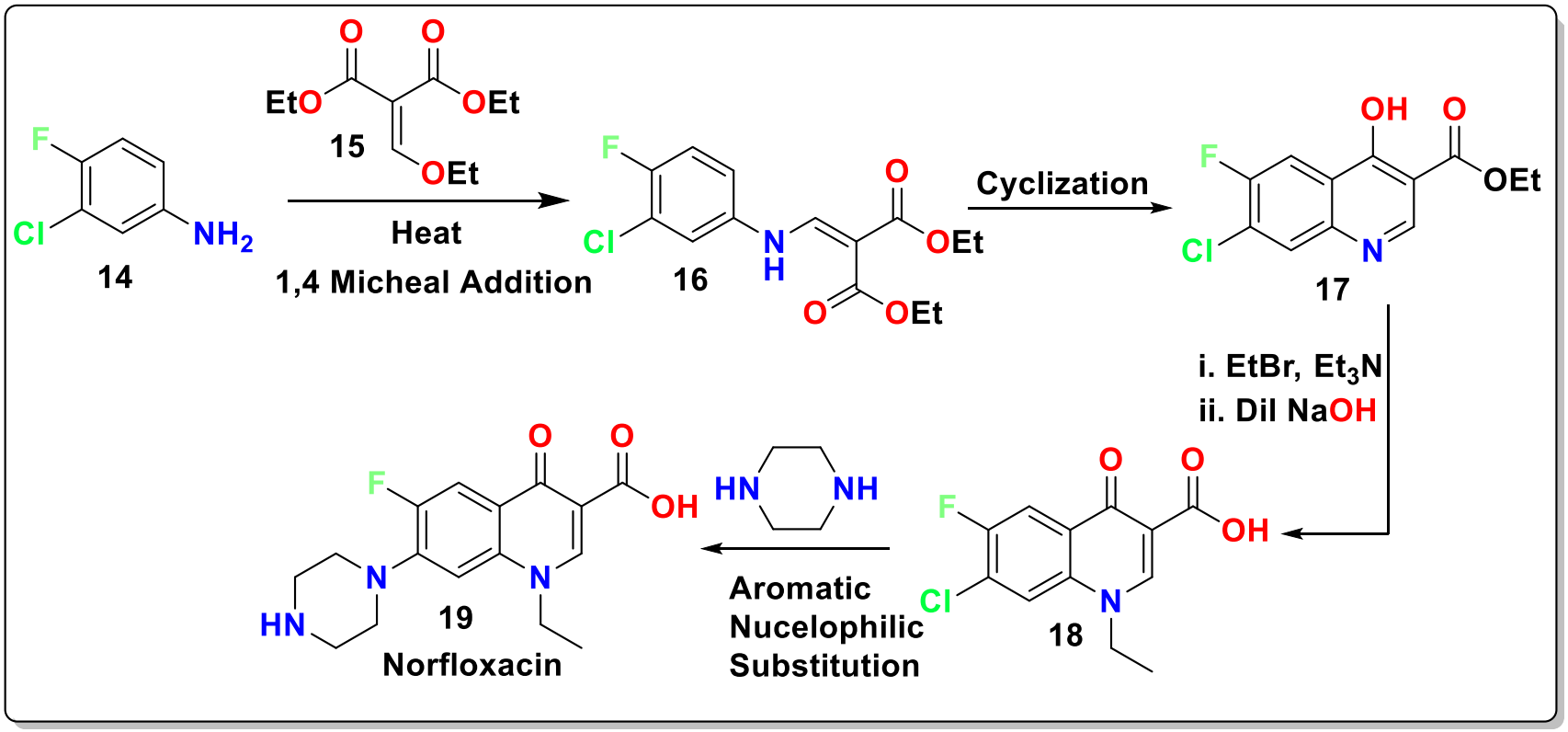
mixture of methyl 2-deoxy-α/β-D-glucopyranoside (24 g) (from example 2) water (125 ml) and IR 120H+resin (7.5 ml) was heated to 90-95° C. for 2 h. The reaction mixture was cooled to room temperature, filtered and the resin was washed with water (20 ml). The filtrate was concentrated to residue and crystallized from ethanol to obtain 2-deoxy-D-glucose (8.8 g), mp. 150-152° C.CLIP

2-deoxy-D-glucose (V)

Norfloxacin

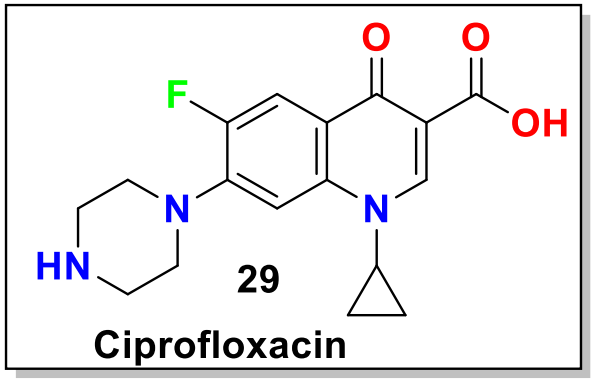
- Norfloxacin is the first representative of a series of fluorinated quinolones as well as the first drug of the quinolone derivatives used in medicine that contains a piperazine substituent.
- It is a broad-spectrum fluoroquinolone antibacterial agent used in the treatment of urinary tract infection.

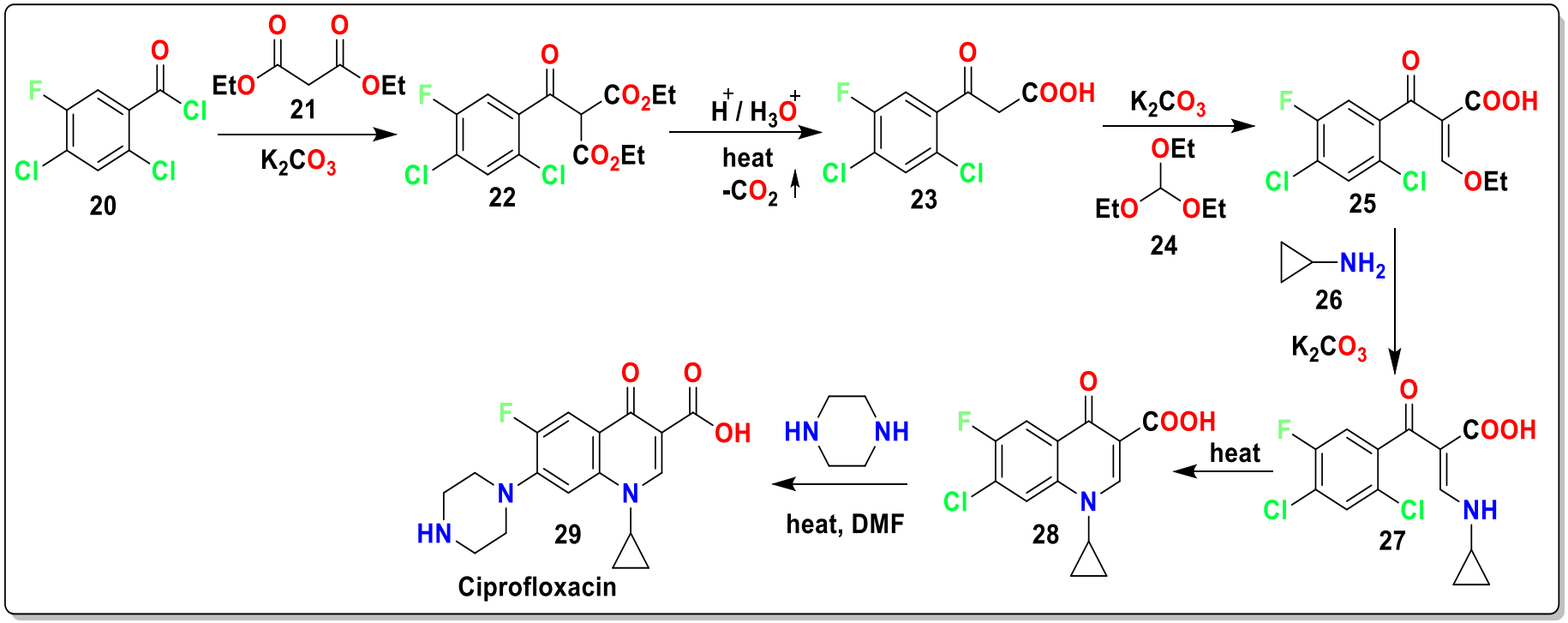


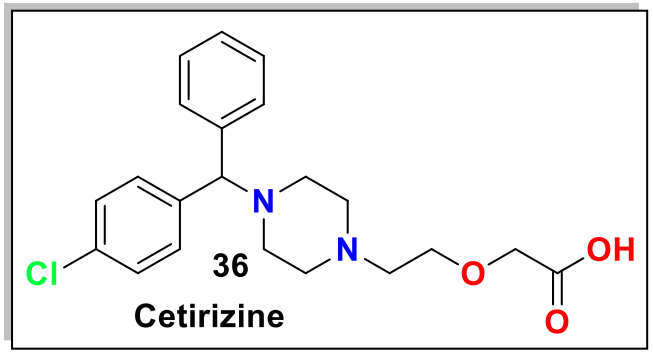
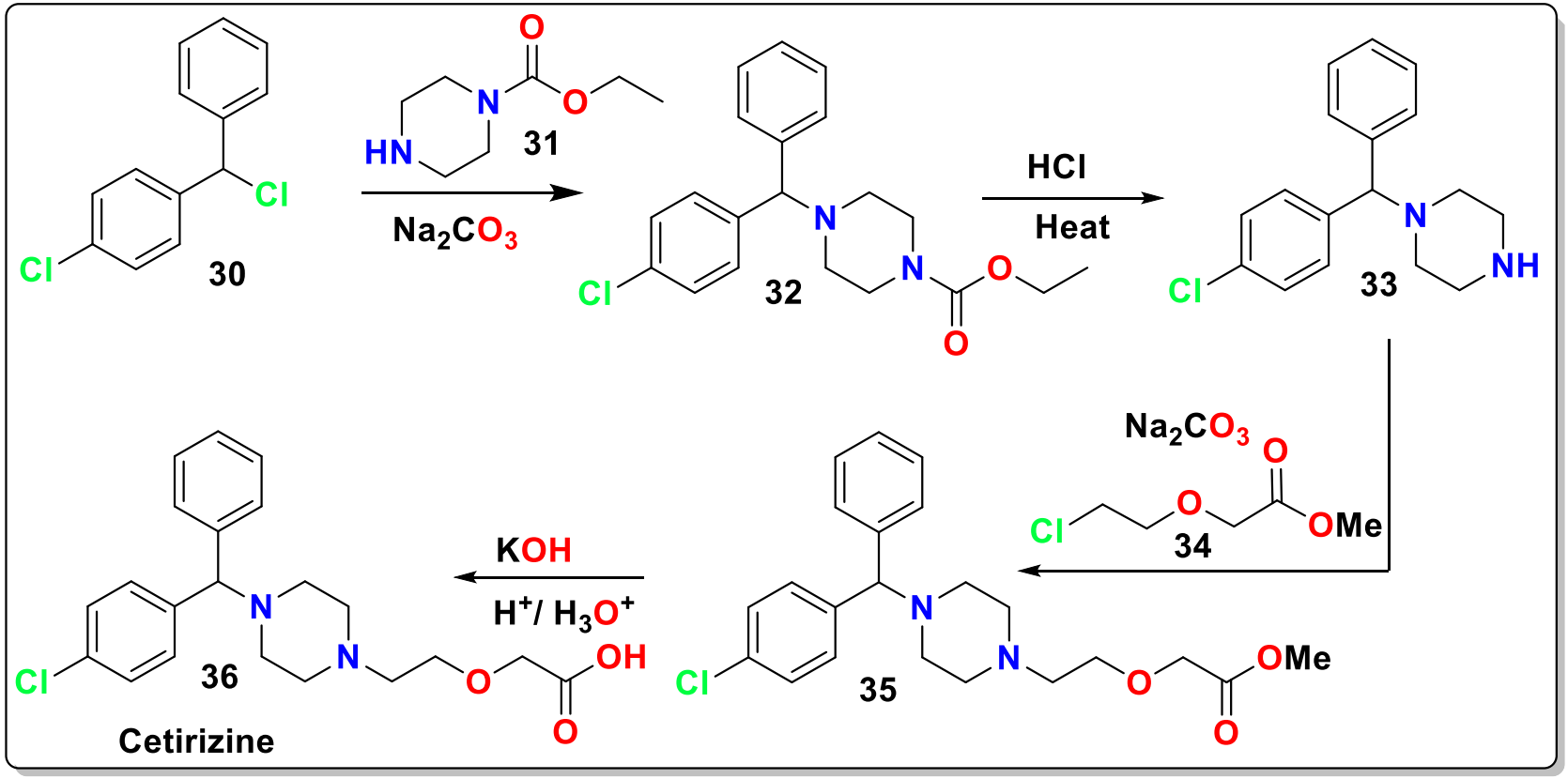


Ciprofloxacin

- Ciprofloxacin is used to treat a wide variety of infections.
- Ciprofloxacin is broad based antibiotic of the fluoro-quinoline class.
- It is active against both Gram-positive and Gram-negative bacteria.





Cetirizine

- Cetirizine is a piperazine derivative and used to treat allergies.